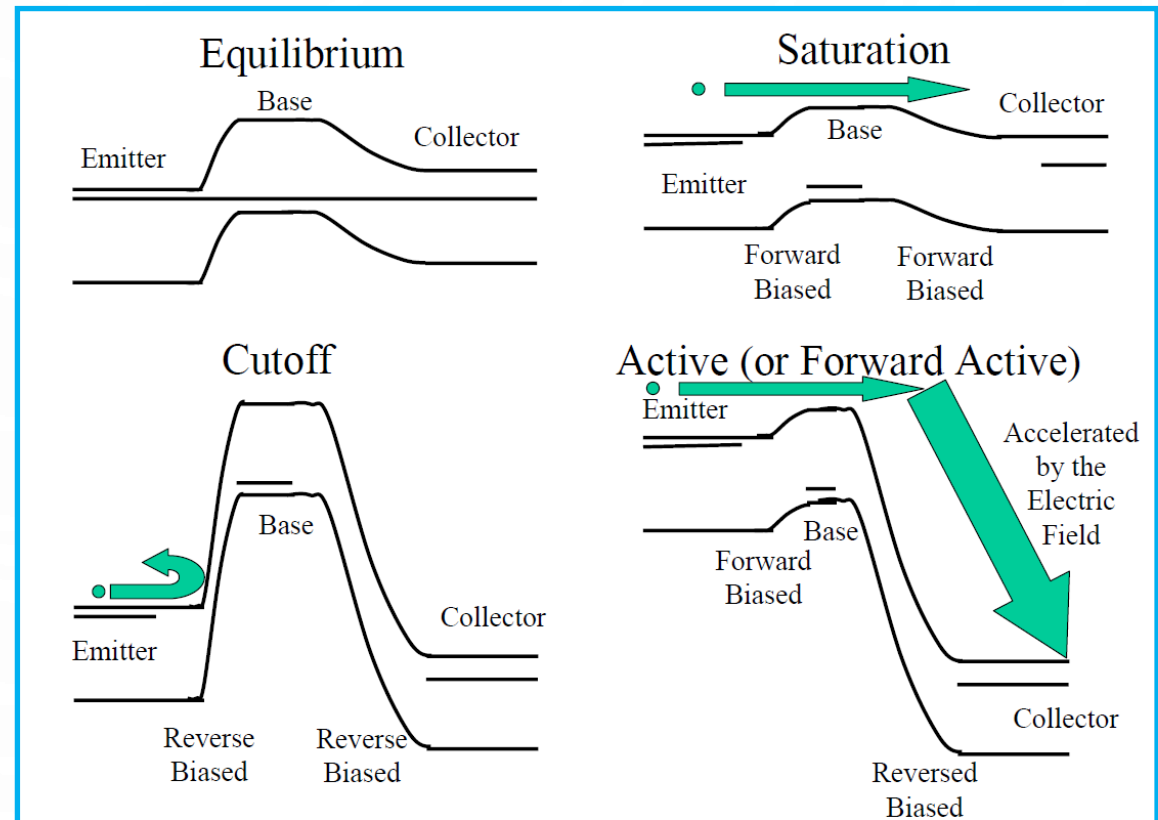
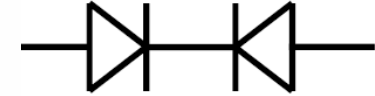
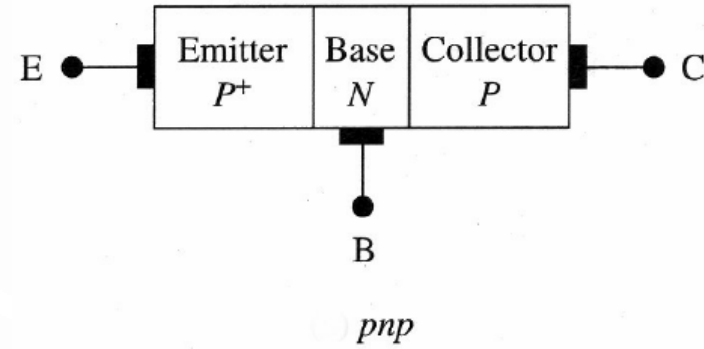
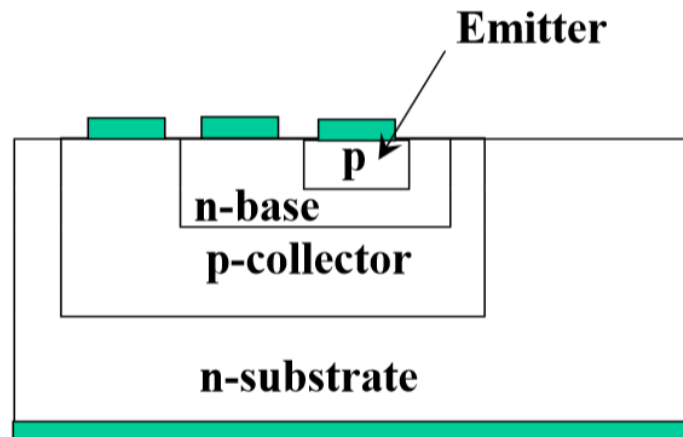
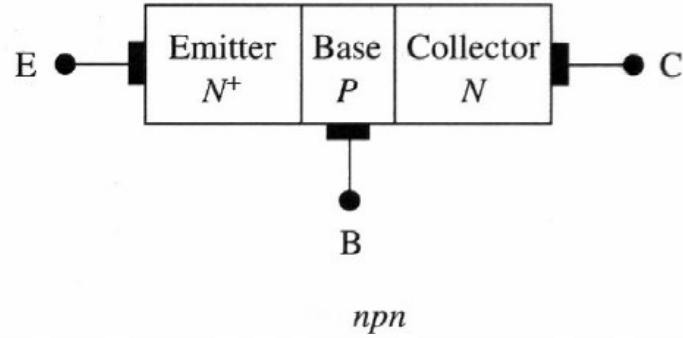


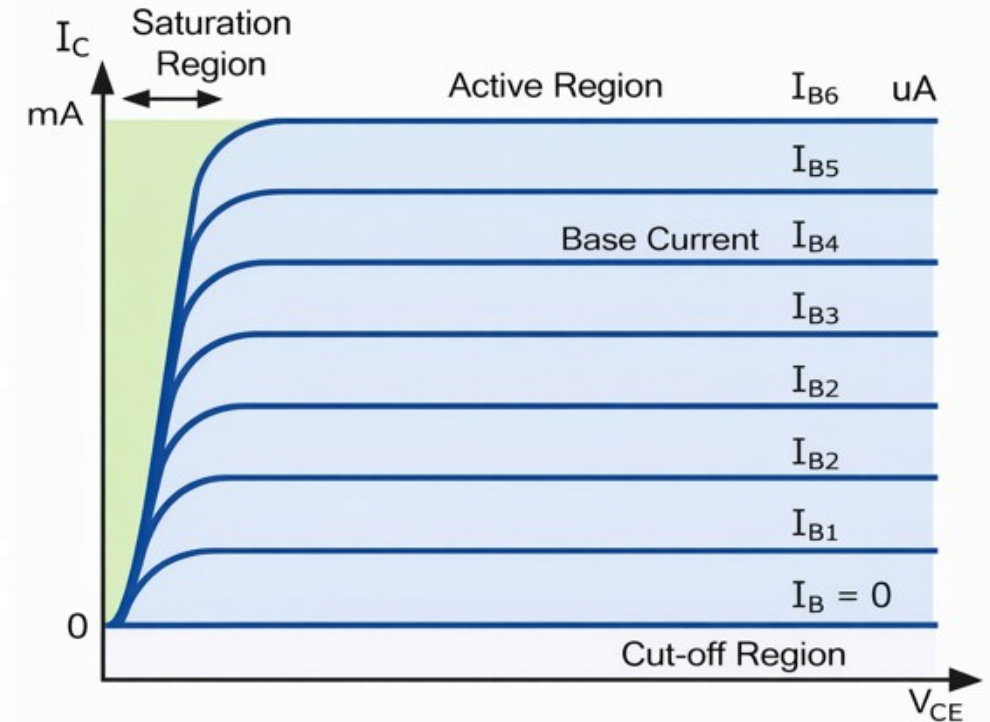
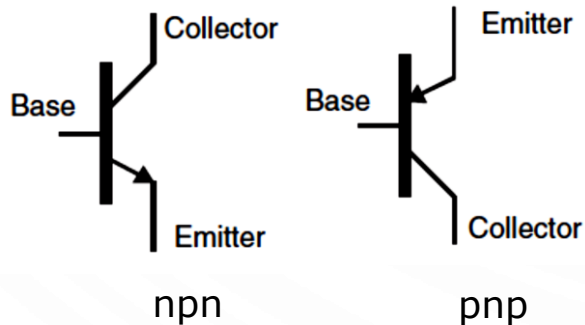
Diodes and Transistors

BJT



BJT

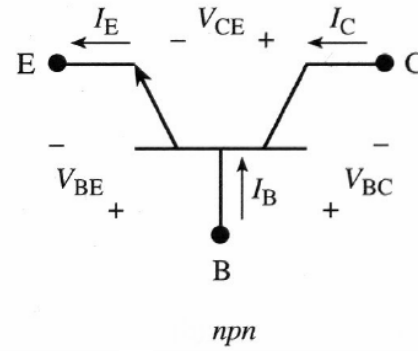
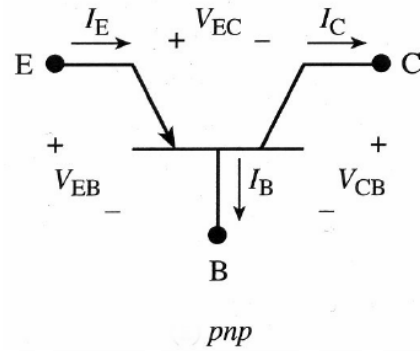
- BJT is a current controlled device
- Has a large blocking voltage in the off state
- Has a high current carrying capability in the on state
- Low current gain => Need high base current to obtain reasonable I_C
- Input impedance is very low => input current is microamps or milliamps in order
- Current-driven => longer switching times
- Expensive and complex base drive circuit => not popular in new products



DC Characteristics of npn BJT

P_{max} is a measure of how much power dissipation the transistor can withstand without any damage due to overtemperature

BJT



$$I_E = I_B + I_C$$
$$V_{EB} + V_{BC} + V_{CE} = 0$$

Forward Active Region

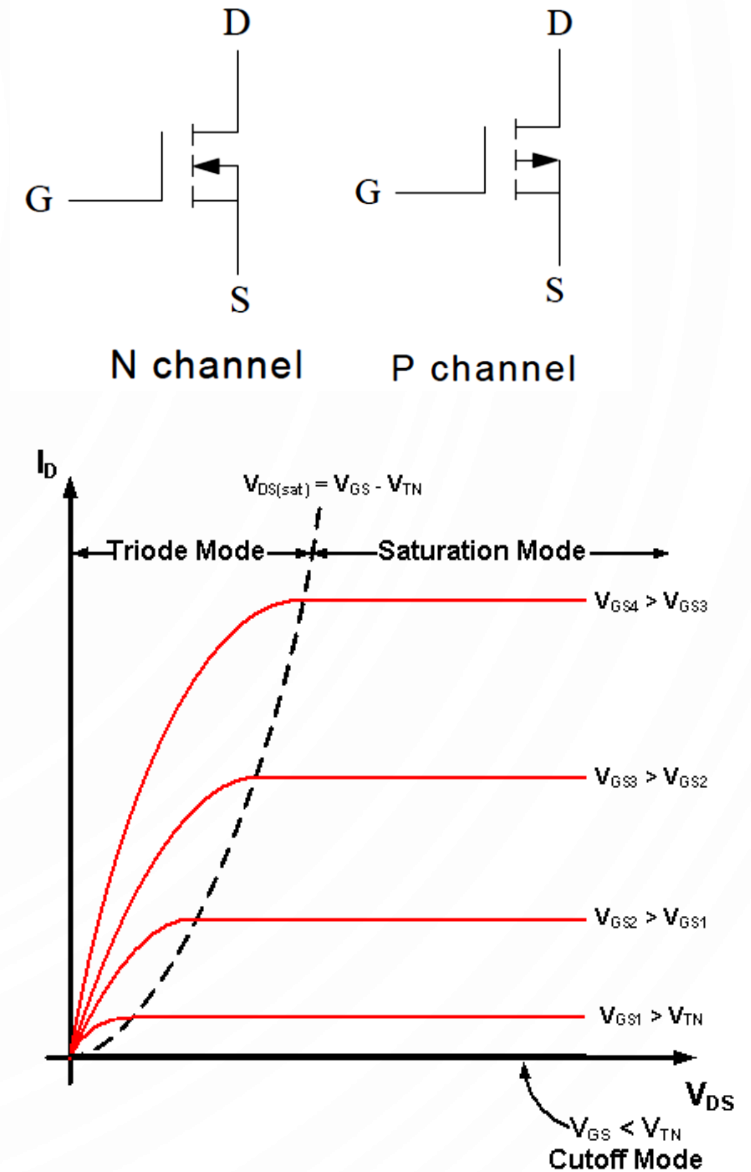


Current Gain

$$\beta = \frac{I_C}{I_B}$$

MOSFET

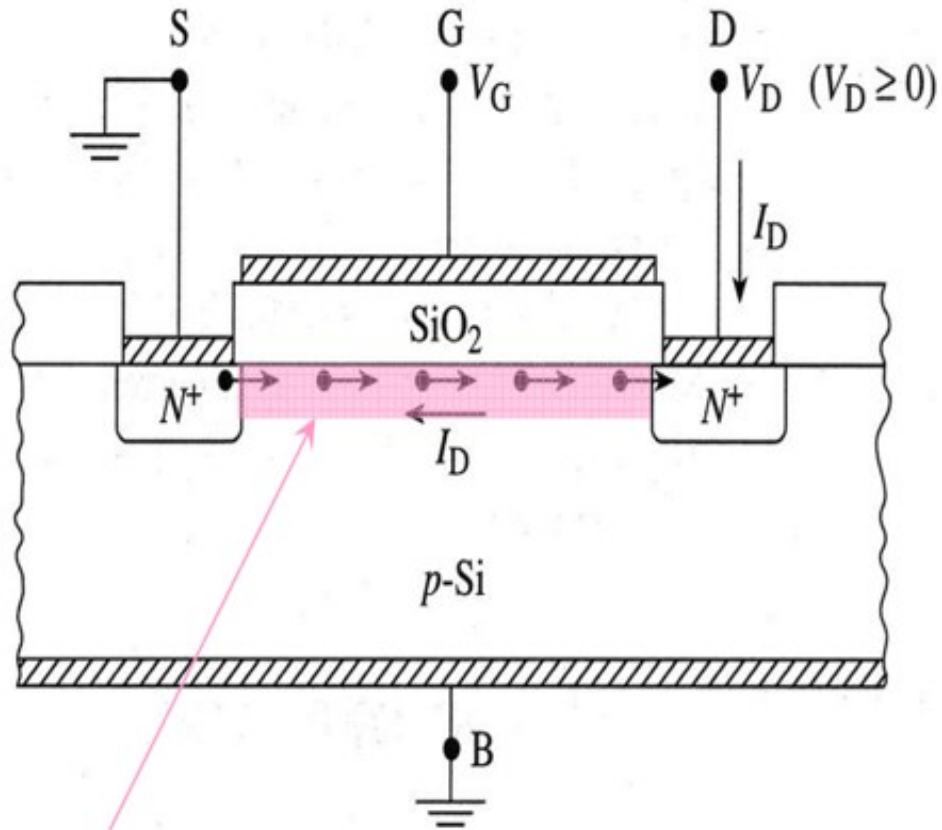
- MOSFET is a voltage controlled device
- Turned 'ON' when a voltage is applied to GS
- Turned 'OFF' by removing the GS voltage
- Voltage driven => faster switching speed => are preferred in applications such as choppers and inverters
- Only voltage drive is required => the drive circuits of MOSFET are very simple
- Have high on-state resistance => for higher currents; losses in the MOSFET's are increased => are used for low/medium power applications
- Input impedance is very high (more than 10^7 Ohms) => input current is approximately zero (picoamps)



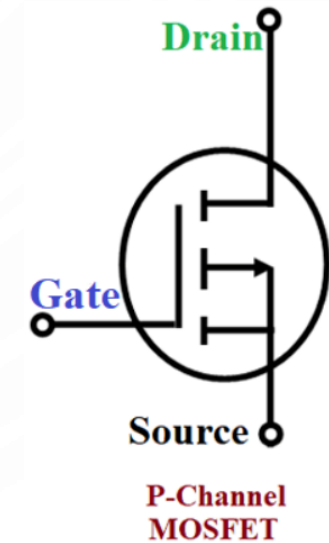
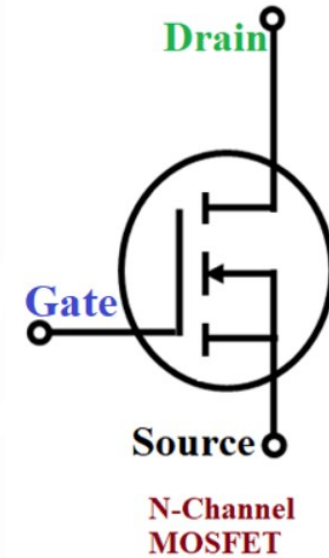
DC Characteristics of n-channel MOSFET

MOSFET

Flow of current source from Source to Drain is controlled by the Gate voltage

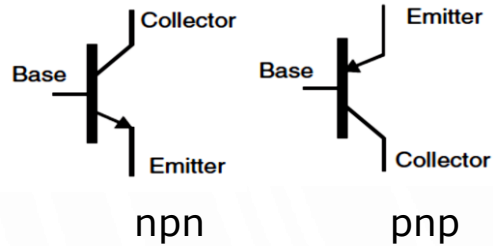


Control by the Gate voltage is achieved by modulating the conductivity of the semiconductor region just below the gate. This region is known as the channel

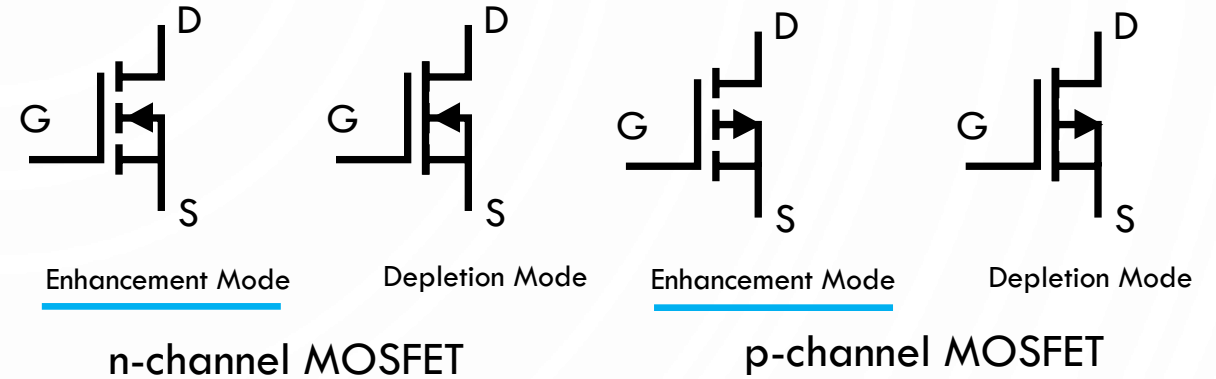


TRANSISTOR SYMBOLS

Bipolar Junction Transistors (BJT)



Metal Oxide Field Effect Transistors (MOSFET)



Enhancement-mode MOSFETs are more commonly used in modern electronics!!!

An enhancement-mode MOSFET is normally OFF and turns ON only when a sufficient gate-source voltage is applied.

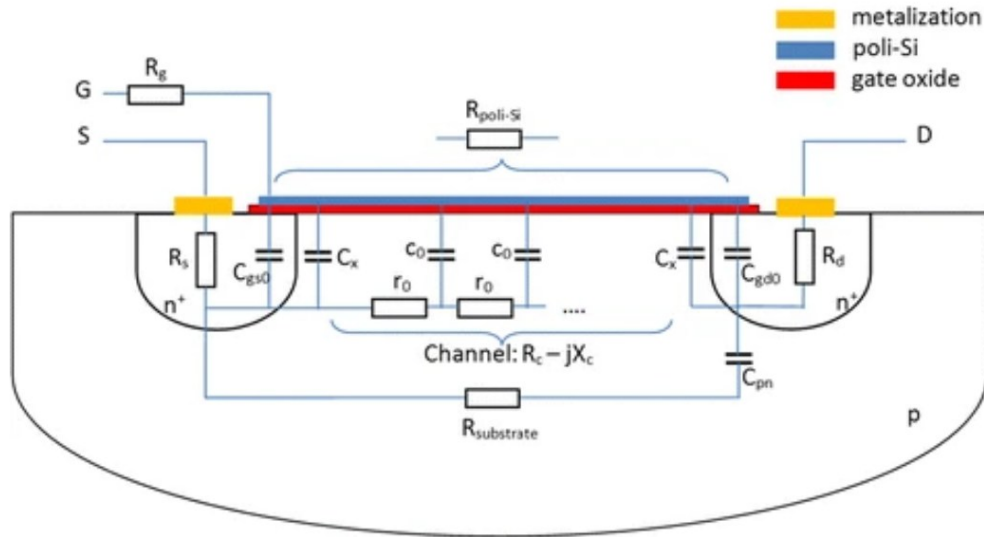
A depletion-mode MOSFET is normally ON and requires a gate-source voltage of opposite polarity to turn it OFF.



Simplified MOSFET symbols

MOSFET

For sufficiently small signals, a MOSFET with parasitics can be characterized as follows:



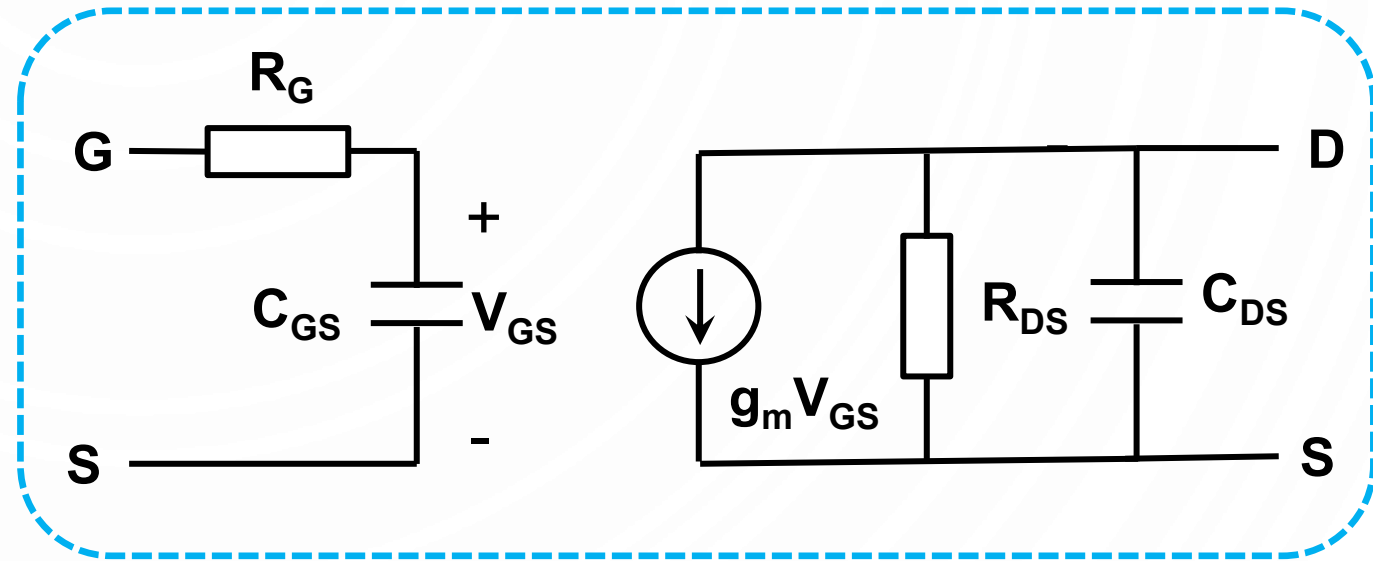
Simplified cross-section of MOSFET

Transconductance in saturation:

$$g_m = 2K (V_{GS} - V_{th})$$

or

$$g_m = 2\sqrt{KI_D}$$



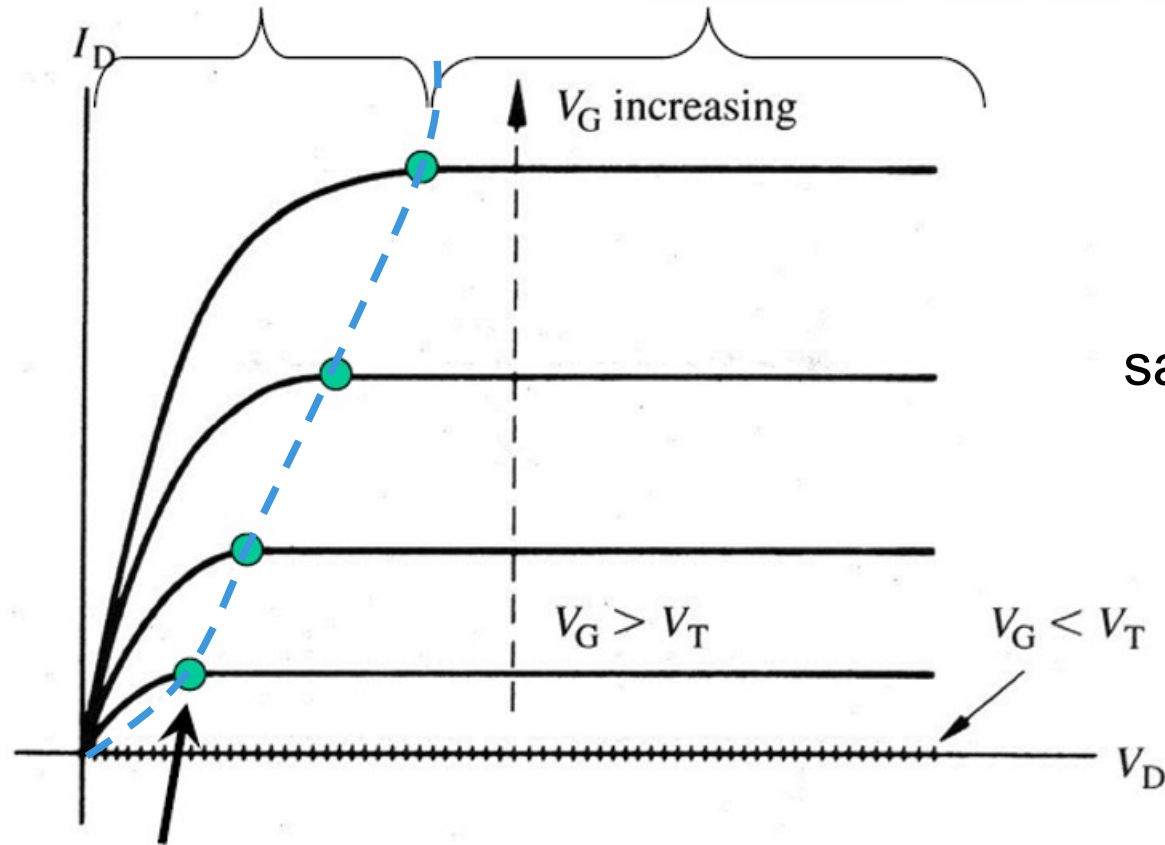
MOSFET simplified small-signal equivalent circuit valid for saturation region. Parasitic components like R_d and R_s are neglected.

MOSFET

$$I_D = \frac{W\mu_n C_{ox}}{L} \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$$
$$0 \leq V_{DS} \leq V_{Dsat} \quad \text{and} \quad V_{GS} \geq V_T$$

$$I_D = I_{Dsat} = \frac{W\mu_n C_{ox}}{2L} (V_{GS} - V_T)^2 \quad V_{Dsat} \leq V_{DS}$$

linear region



saturation region

$$V_{Dsat} = V_{GS} - V_T$$

MOSFET

MOSFETS have usually two applications:

- Switches, inverters, logic gates
- Amplifiers

If a MOSFET is used as a switch, the MOSFET is operated in the linear region.

If a MOSFET is used in an amplifier, the MOSFET is operated in the saturation region.

MOSFET

Linear Region

When the MOSFET is fully enhanced and driven hard on:

$$V_{GS} \gg V_{TH}$$

V_{DS} is forced small by the external circuit

This places the device in the triode (linear / ohmic) region, where:

$$V_{DS} \ll V_{GS} - V_{TH}$$

This is ideal for a switch:

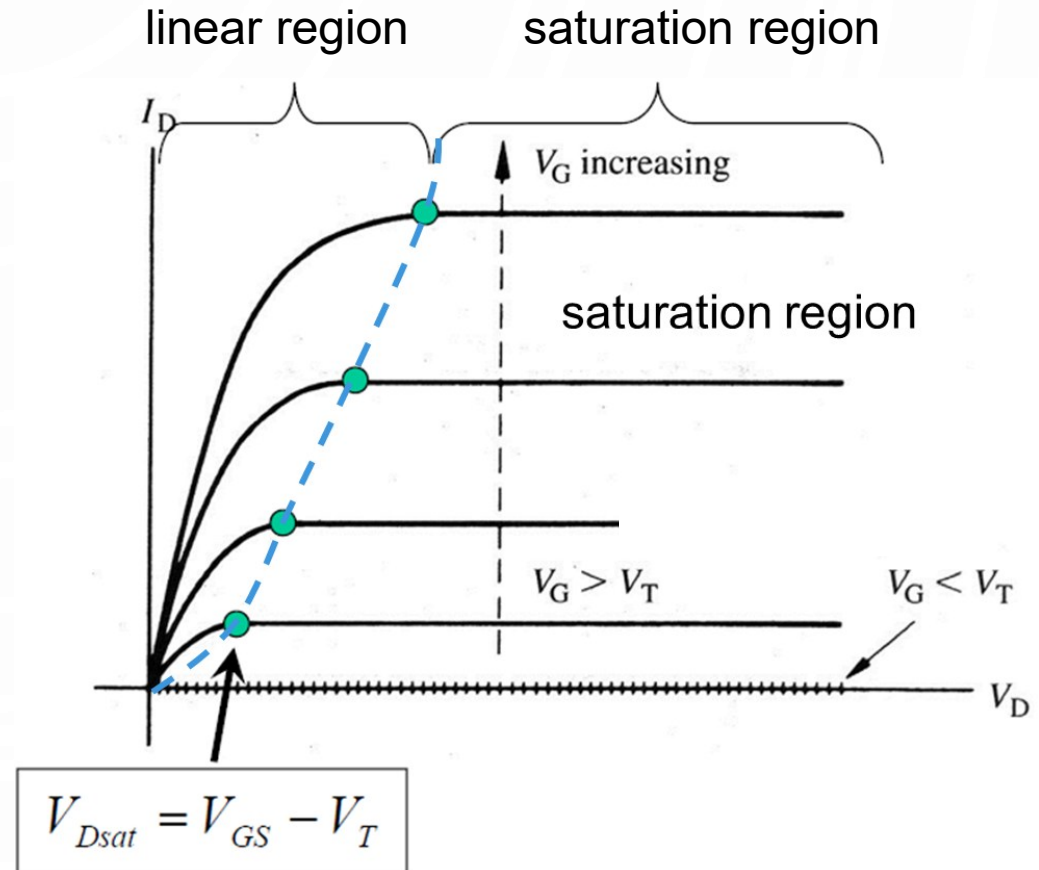
- Low $R_{DS(on)}$
- Low conduction loss $P \approx I_D^2 R_{DS(on)}$
- Minimal heat
- High efficiency

If a switch were operated in saturation:

- V_{DS} would be relatively large
- Power loss $P = V_{DS} \cdot I_D$ would be high
- Efficiency would be poor
- Thermal stress would increase

$$I_D = \frac{W\mu_n C_{ox}}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$
$$0 \leq V_{DS} \leq V_{Dsat} \quad \text{and} \quad V_{GS} \geq V_T$$

$$I_D = I_{Dsat} = \frac{W\mu_n C_{ox}}{2L} [(V_{GS} - V_T)^2] \quad V_{Dsat} \leq V_{DS}$$



MOSFET

Saturation Region

$$V_{GS} > V_{TH} \text{ and } V_{DS} \geq V_{GS} - V_{TH}$$

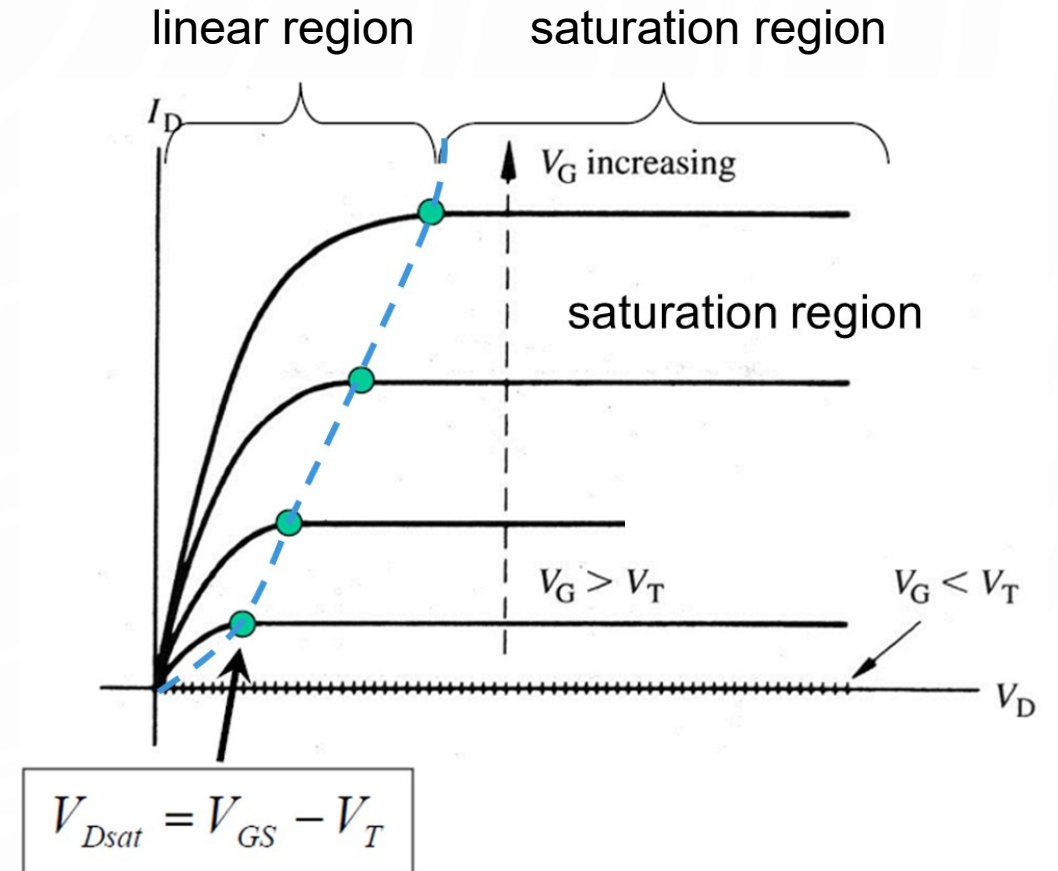
Channel is pinched off near the drain; drain current is mainly controlled by V_{GS} .

Why amplifiers use saturation region:

In amplification, we want the drain current I_D to be primarily controlled by the gate-source voltage V_{GS} , and to be relatively insensitive to the drain-source voltage V_{DS} .

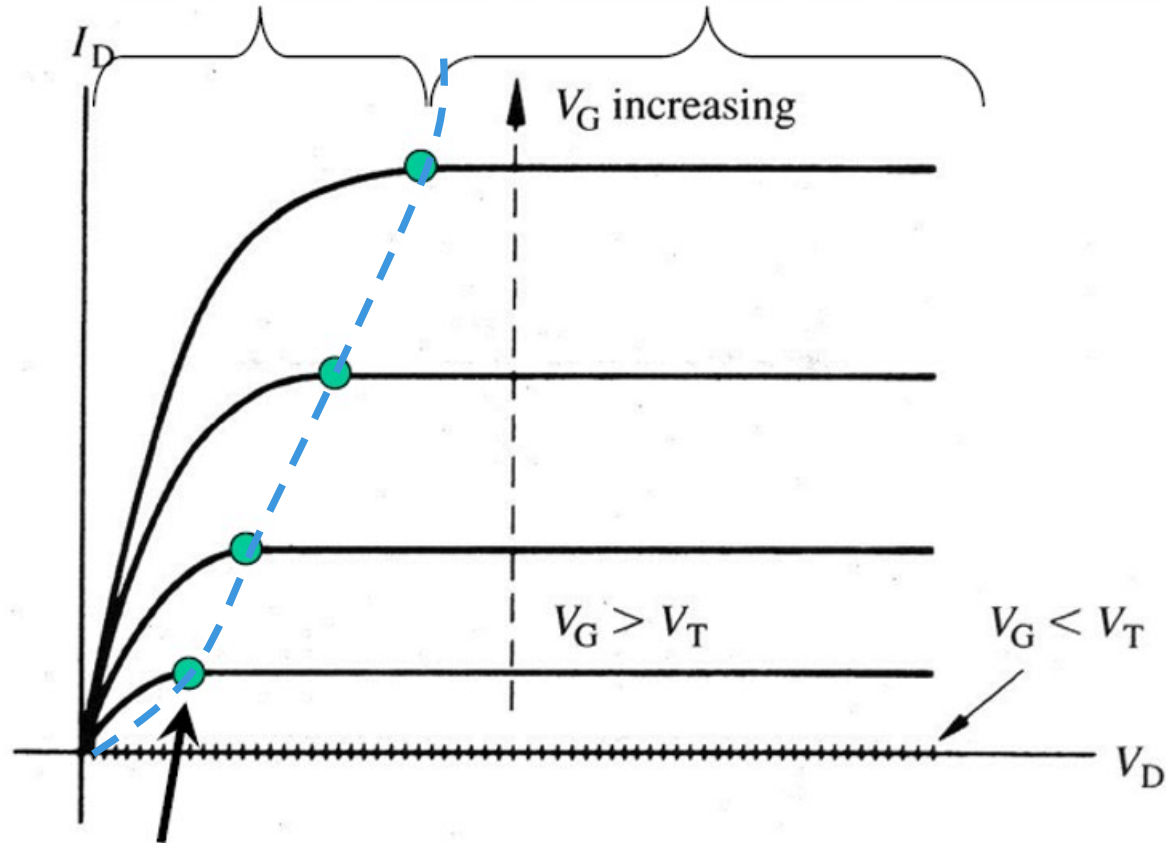
This gives:

- High transconductance $g_m = \partial I_D / \partial V_{GS}$
- A well-defined, approximately quadratic or linearized relationship between input (V_{GS}) and output current
- The ability to convert small changes in V_{GS} into proportional changes in I_D and hence output voltage via a load



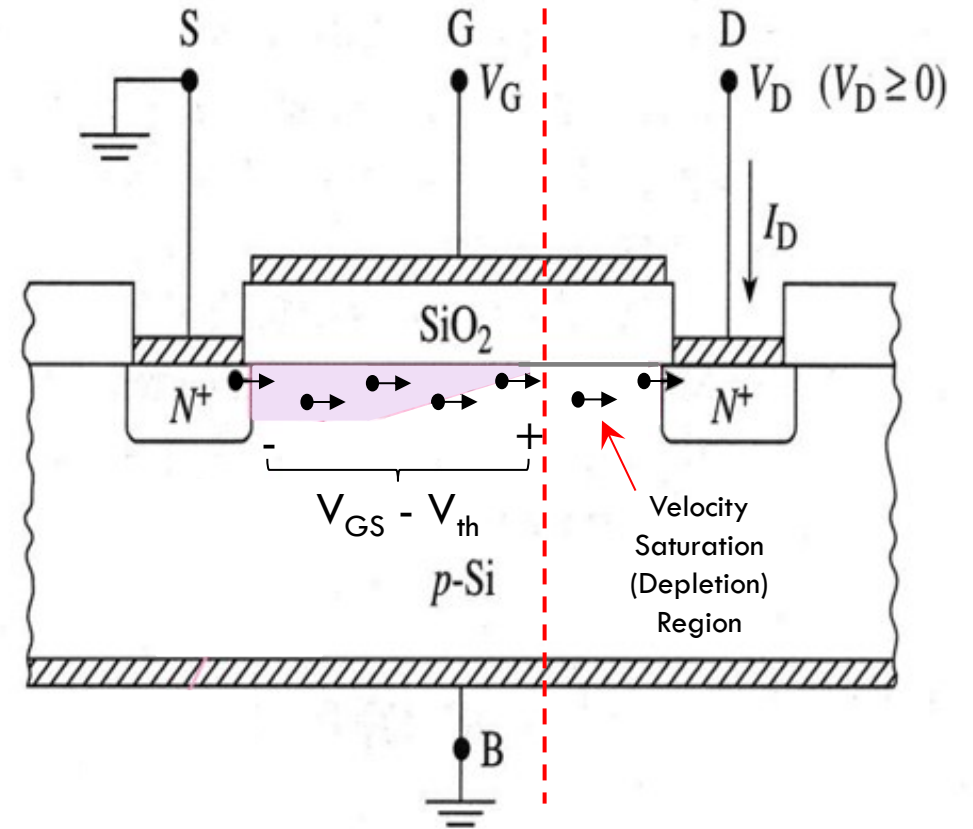
MOSFET

CHANNEL LENGTH MODULATION



$$V_{Dsat} = V_{GS} - V_T$$

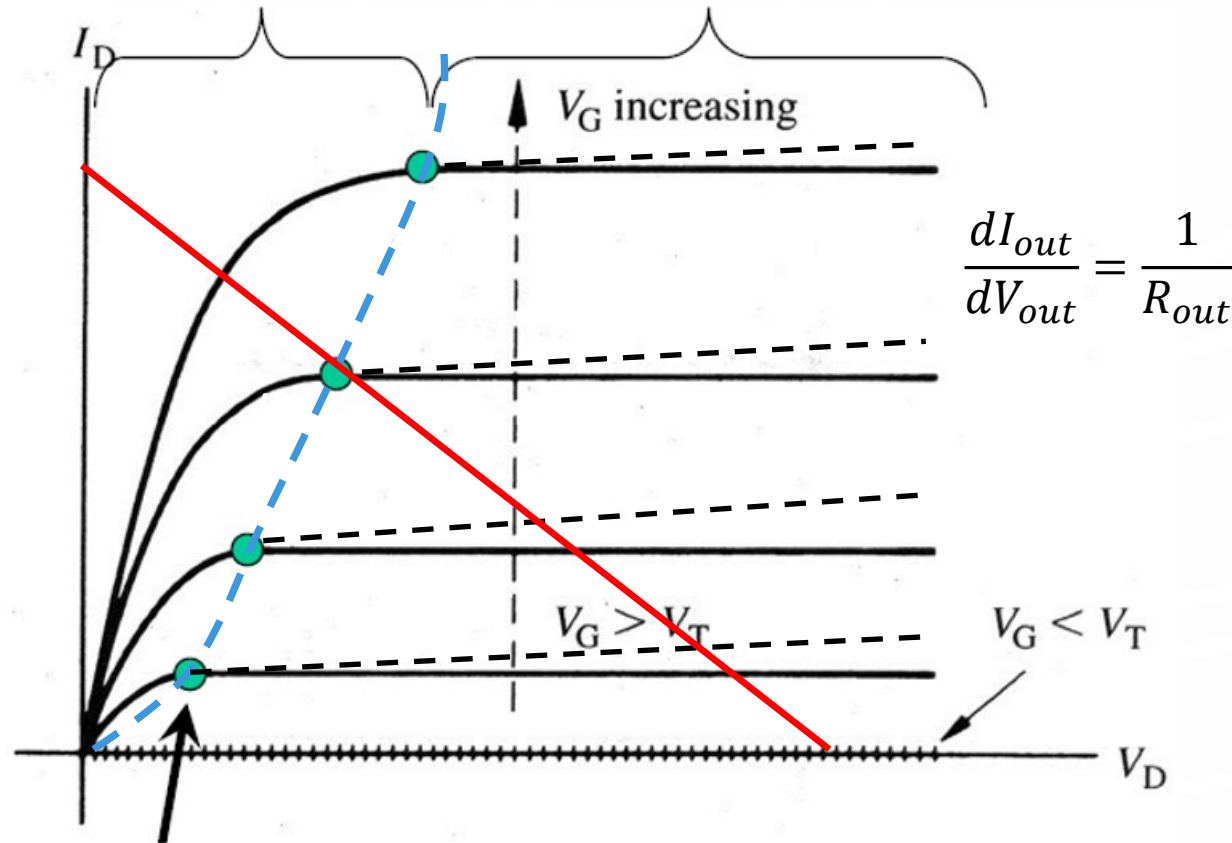
R_{out} , which is the reciprocal of the slope in the correct operation zone, must be as large as possible



When the Drain-Source voltage reaches a particular value, the induced charge near the drain becomes zero and the channel gets *pinched-off*. The voltage difference between the pinch-off point and the source remains constant even if V_{DS} is increased. Thus, the current flow I_D saturates, and the transistor is said to be in *saturation*.

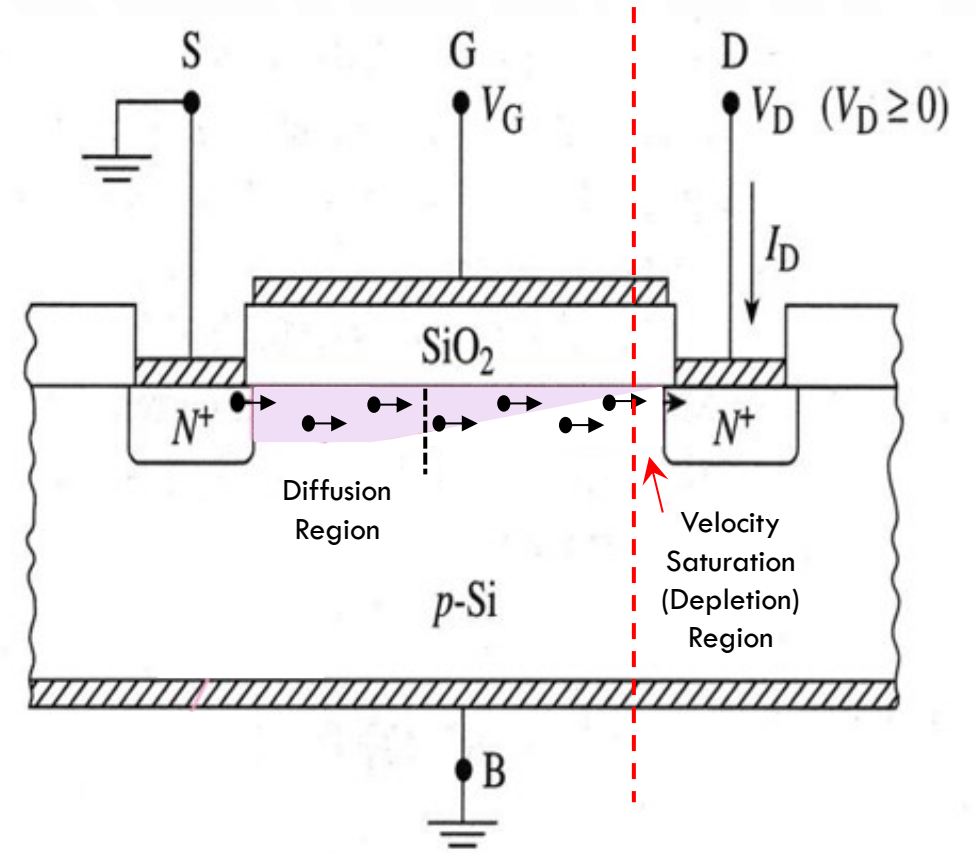
MOSFET

CHANNEL LENGTH MODULATION (dashed-lines)



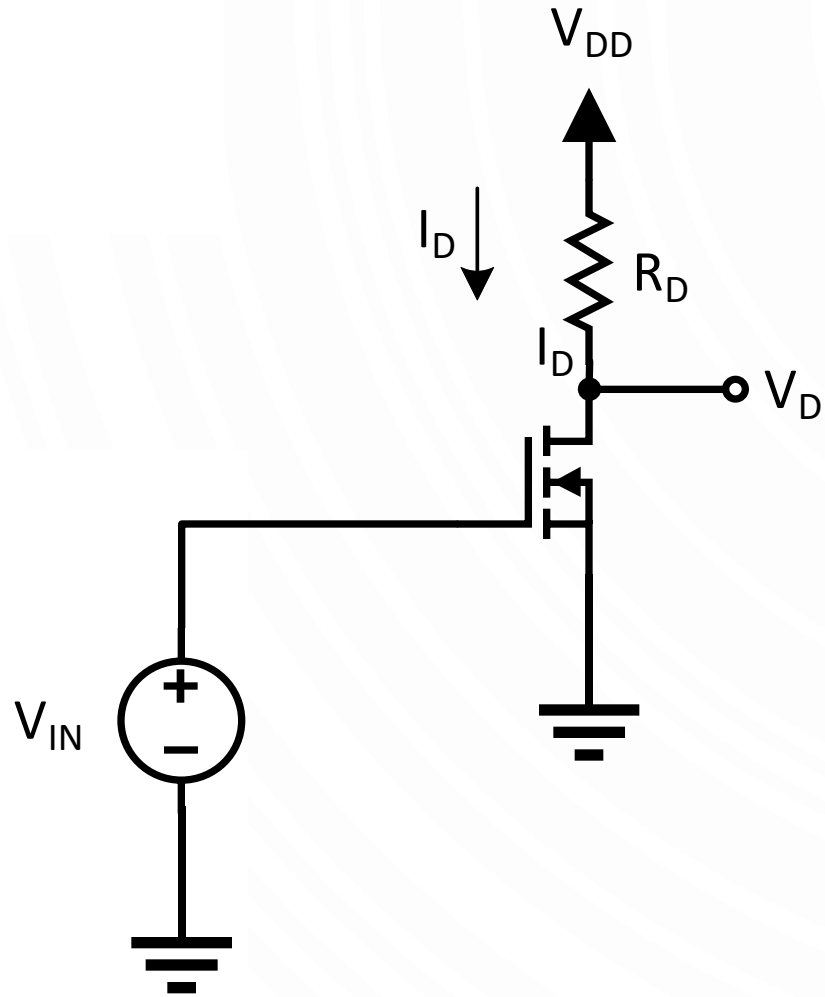
$$V_{Dsat} = V_{GS} - V_T$$

R_{out} , which is the reciprocal of the slope in the correct operation zone, must be as large as possible



A depletion region is formed at the drain side of the channel at the n^+ and p junction. Charge carriers (electrons) move to the drain due to the electric field. Within the depletion region, electrons move at a constant velocity (velocity saturation). Effective gate length becomes smaller due to the depletion region → channel length modulation

SIMPLE APPLICATION EXAMPLES

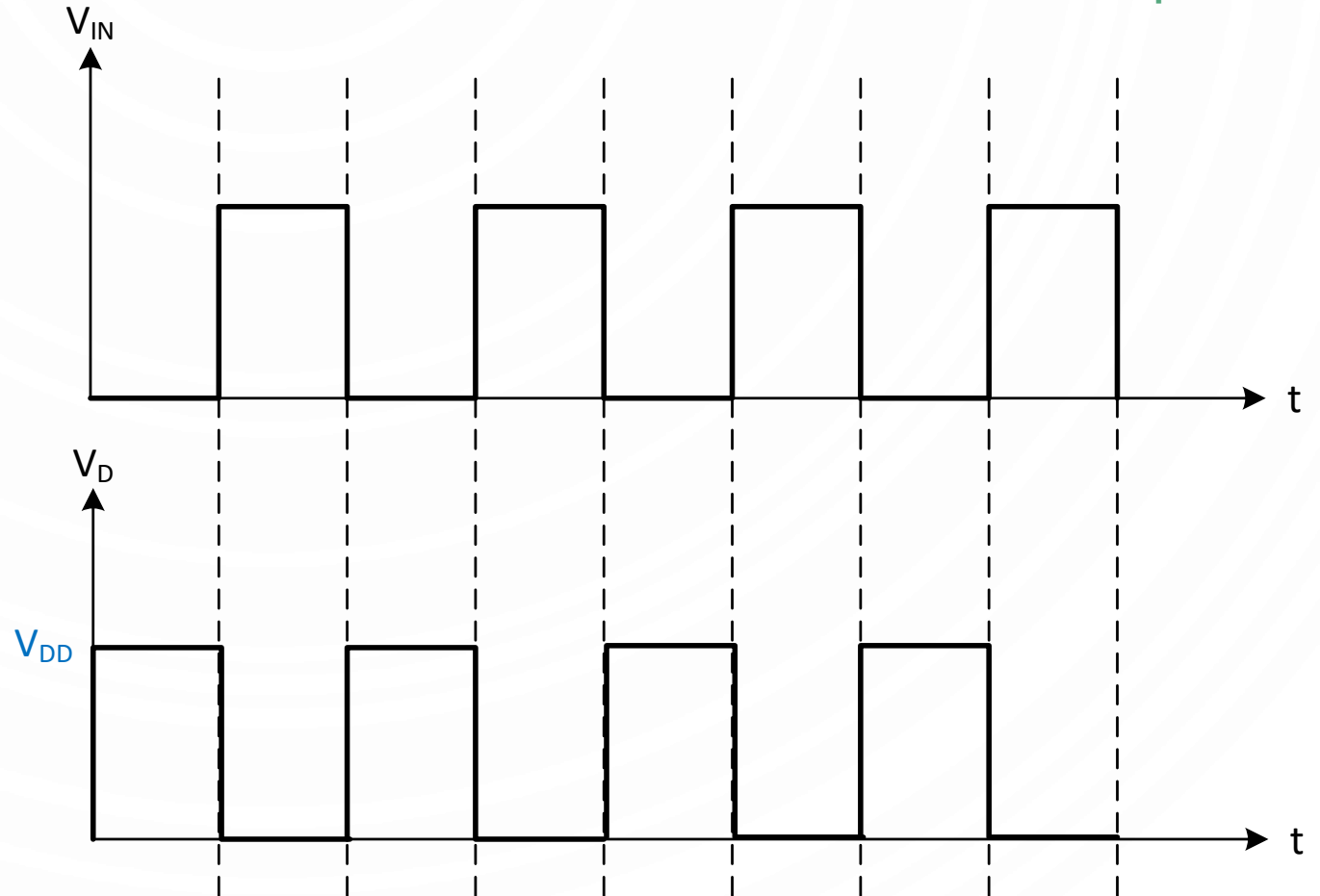


MOSFET acting as a Switch

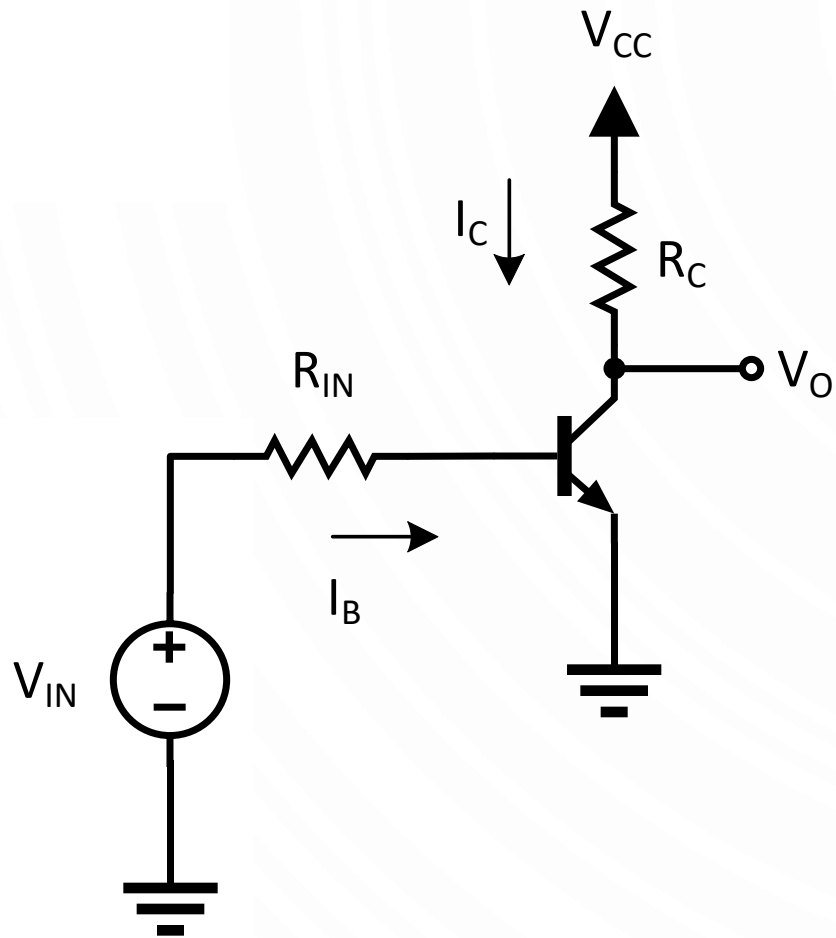
$$V_D = V_{DD} - R_D I_D$$

$$V_{IN} \gg V_{GS,TH}$$

180° Phase Shift between Input and Output



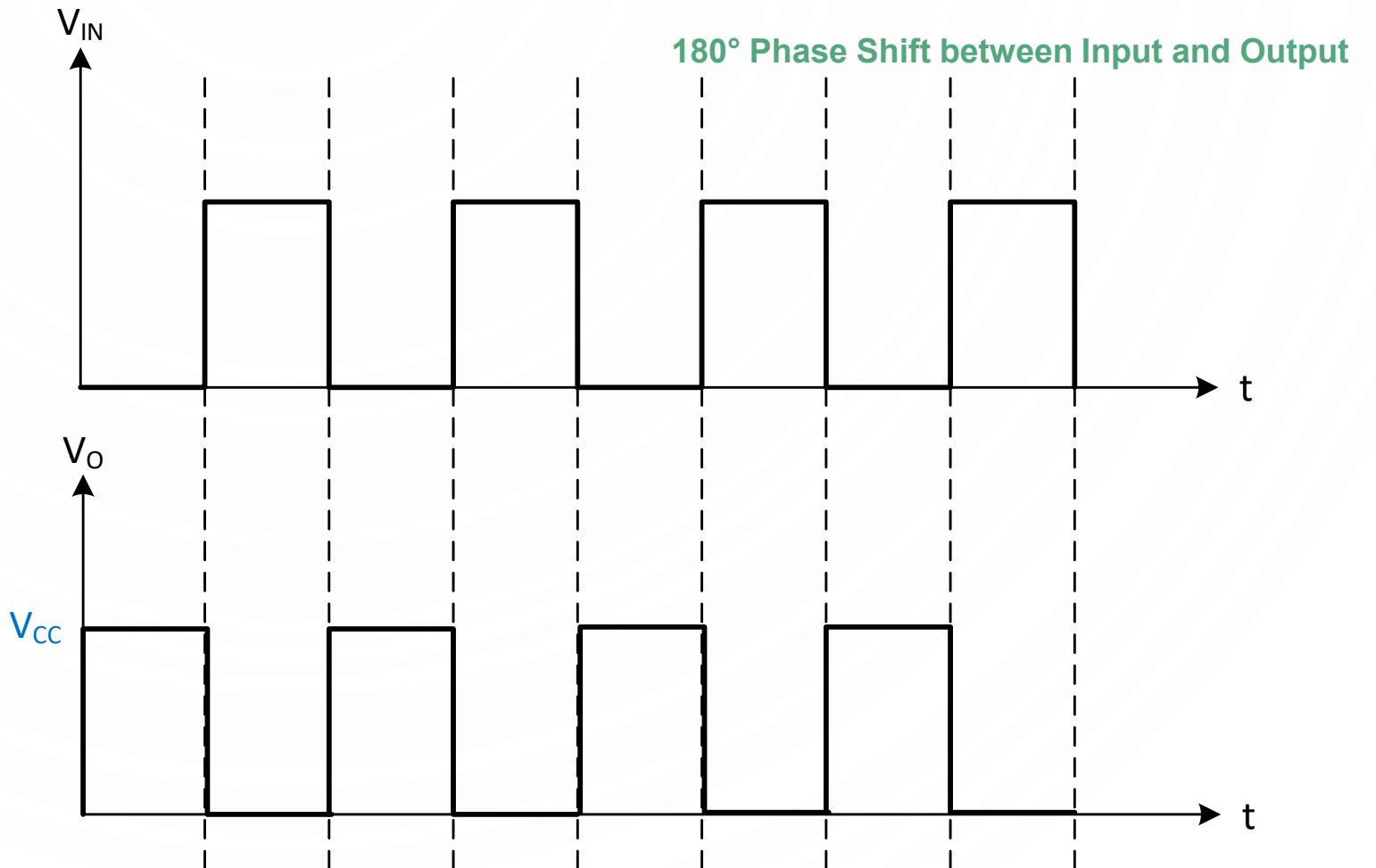
SIMPLE APPLICATION EXAMPLES



BJT acting as a Switch

$$V_O = V_{CC} - R_C I_C$$

V_{IN} produces sufficient base current through R_{IN} to drive the transistor into saturation



References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.